CAVS OFF-ROAD PROVING GROUND · AUTONOMOUS VEHICLE TESTING

Supervisory Monitor

Dashboard sketches · v2

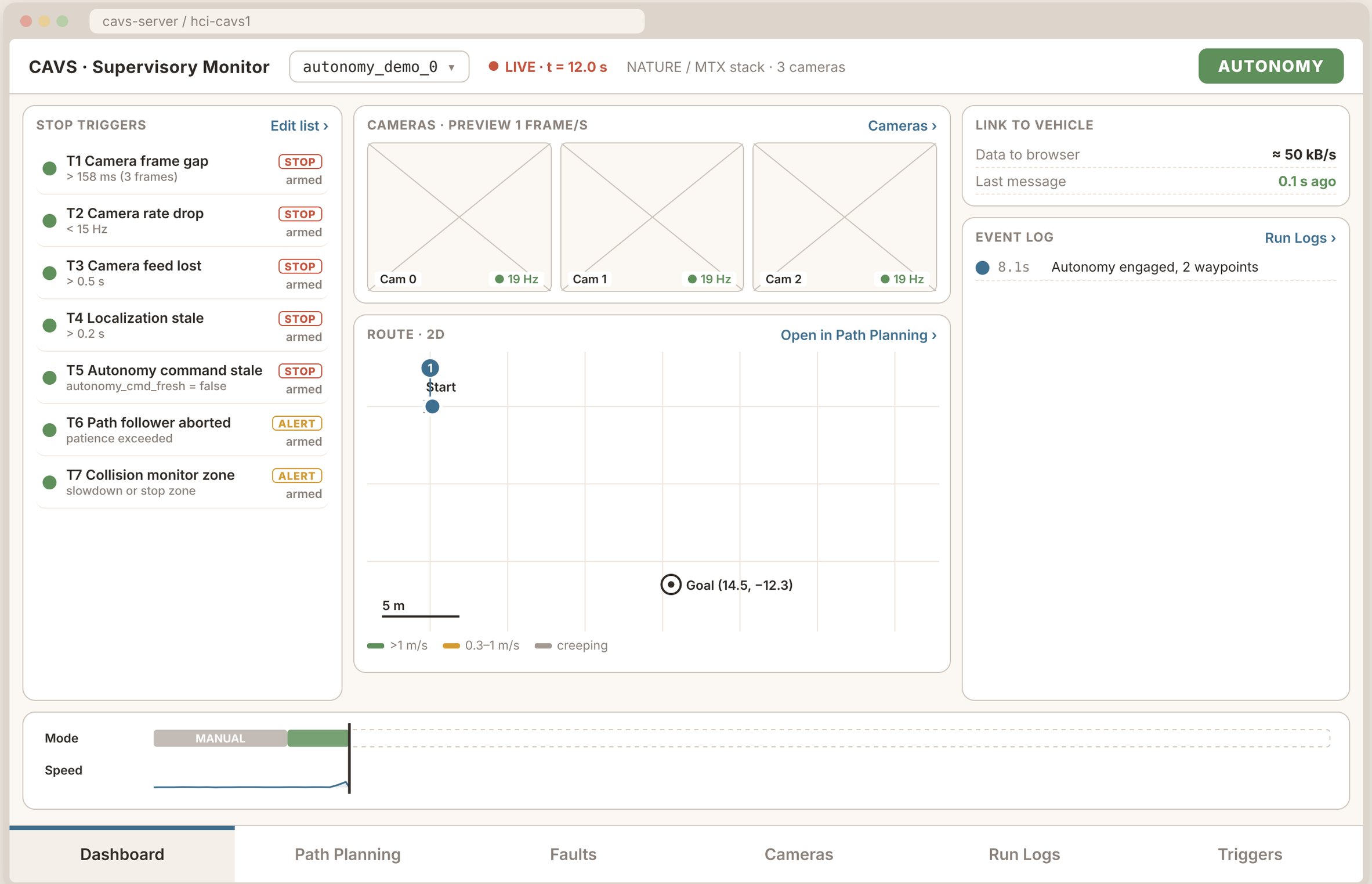
Use case	Watch an autonomy run and stop the vehicle automatically when a listed trigger fires. The safety driver then takes over. Runs can be reviewed afterwards.
Where it runs	Web interface hosted on a CAVS server, opened in a desktop browser
Sensors in scope	The 3 vehicle cameras. 2D route map only, no 3D view.
Data in these sketches	Replay of the provided bag autonomy_demo_0 (72.1 s, 124,845 messages)
Pages	1 Live, normal · 2 Live, trigger fired · 3 Run review · 4 Trigger list · 5 Data budget · 6 Navigation

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1 · Dashboard, live — normal driving

Replay of autonomy_demo_0 at 12.0 s. Autonomy active, every trigger armed.



A The mode chip is the first thing read: AUTONOMY, STOPPED or MANUAL.

B Trigger list on the left shows each stop condition and its live state. Green means armed and not met.

C Camera previews refresh once per second. They confirm the feeds are alive without streaming video.

D Route is drawn in 2D from the vehicle pose only.

E No drive controls. A person in the vehicle can always take over.

2 · Dashboard, live — trigger fired

Same replay at 17.9 s. Camera 2 goes 265 ms without a frame, so T1 fires.

cavs-server / hci-cavs1

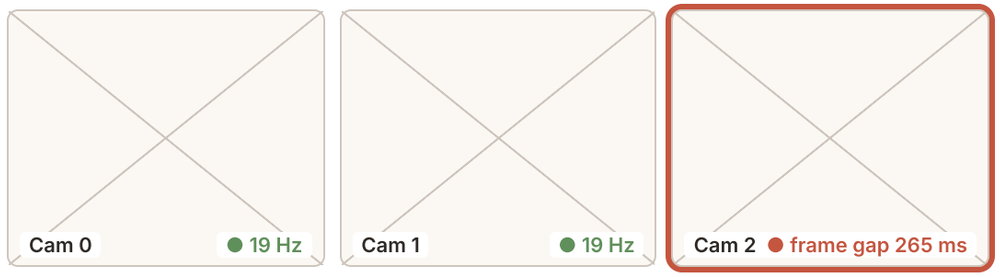
CAVS · Supervisory Monitor autonomy_demo_0 ● LIVE · t = 17.9 s NATURE / MTX stack · 3 cameras **STOPPED**

STOP sent · 17.9 s T1 Camera frame gap: Camera 2, 265 ms (limit 158 ms). Vehicle halted, safety driver in control.

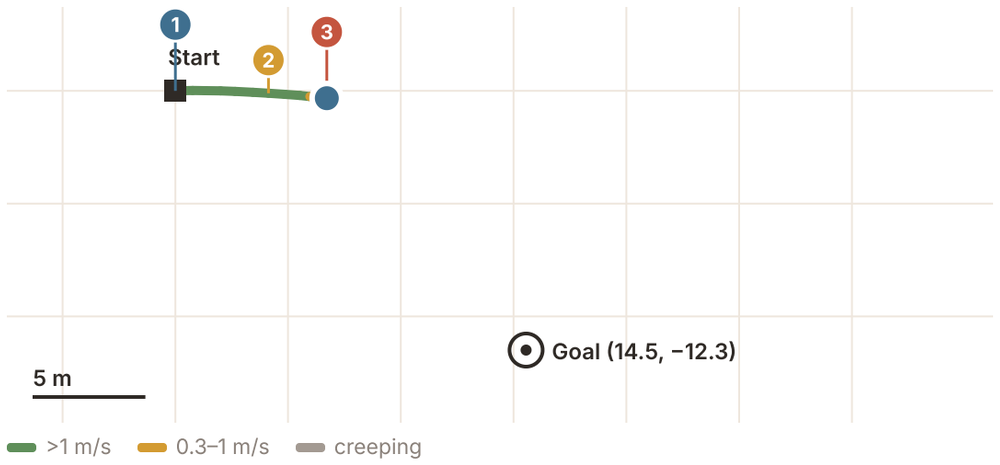
STOP TRIGGERS — Edit list ›

- T1 Camera frame gap — > 158 ms (3 frames) — STOP 265 ms
- T2 Camera rate drop — < 15 Hz — STOP armed
- T3 Camera feed lost — > 0.5 s — STOP armed
- T4 Localization stale — > 0.2 s — STOP armed
- T5 Autonomy command stale — autonomy_cmd_fresh = false — STOP armed
- T6 Path follower aborted — patience exceeded — ALERT at 15.5 s
- T7 Collision monitor zone — slowdown or stop zone — ALERT armed

CAMERAS · PREVIEW 1 FRAME/S — Cameras ›



ROUTE · 2D — Open in Path Planning ›



LINK TO VEHICLE

Data to browser	≈ 50 kB/s
Last message	0.1 s ago

EVENT LOG — Run Logs ›

- 17.9s STOP sent on /mtx/estop · safety driver in control
- 17.9s T1 Camera 2 frame gap 265 ms
- 15.5s T6 Path follower aborted
- 8.1s Autonomy engaged, 2 waypoints



Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

A A full-width banner states which trigger fired, the measured value against its limit, and that the stop was sent.

B The tripped trigger and camera are outlined in red. The other triggers keep their state.

C The stop is sent once and is not repeated. Resuming autonomy is done from the vehicle, not from this screen.

3 · Dashboard, review — after the run

The full run on one time axis, with the trigger list applied to the recorded data.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor autonomy_demo_0 ▾ REVIEW NATURE / MTX stack · 3 cameras MANUAL

REPLAYED AGAINST THE TRIGGER LIST

Actual run	With triggers
64.7 s	17.9 s
operator STOP	T1 Camera 2 frame gap

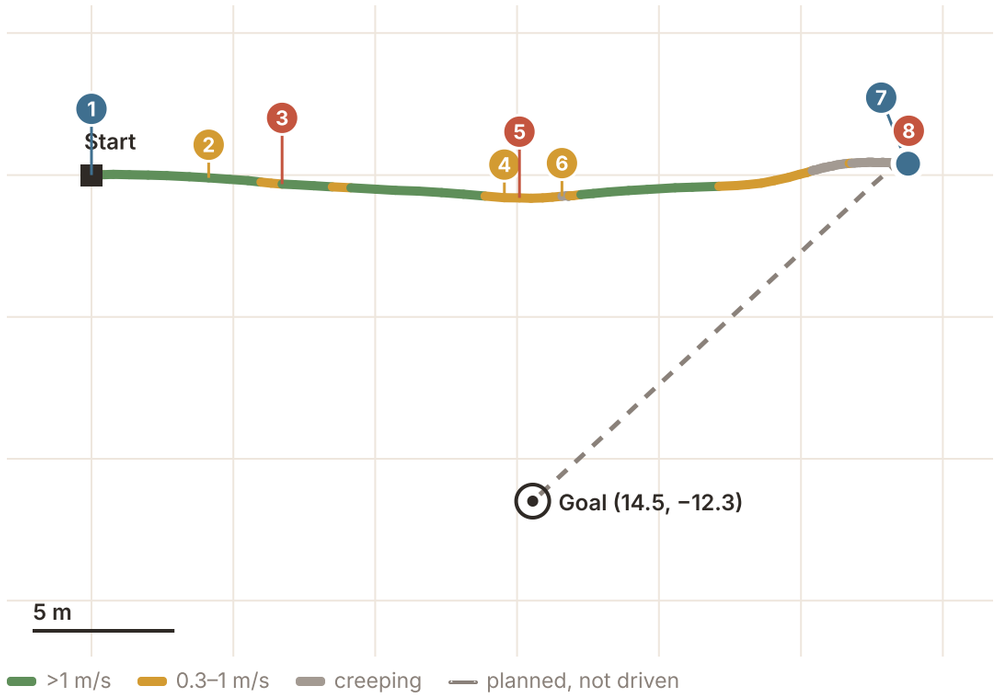
Autonomy that would not have run	46.8 s
STOP conditions met	3 (T1 ×2, T2 ×1)
Alerts	3 (T6 ×2, T7 ×1)

EVENTS

1	8.1s	Autonomy engaged
2	15.5s	T6 Path follower aborted
3	17.9s	T1 Camera 2 frame gap 265 ms
4	23.2s	T7 Slowdown zone
5	23.6s	T2 Camera 2 at 14 Hz
6	25.7s	T6 Path follower aborted
7	64.7s	Operator STOP (actual run)
8	65.8s	T1 Camera 2 frame gap 170 ms

ROUTE · 2D

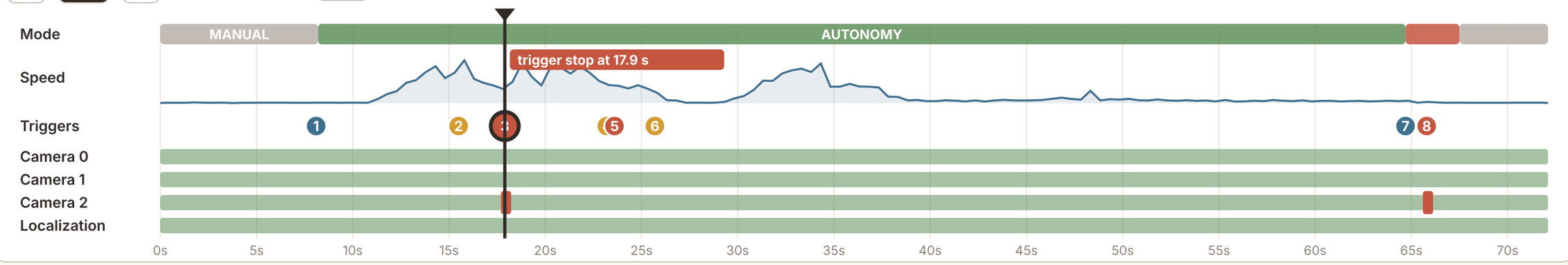
Open in Path Planning ›



FEED HEALTH · WHOLE RUN

Camera 0	/cam_0/image_raw	19.1 Hz · no gaps
Camera 1	/cam_1/image_raw	19.1 Hz · no gaps
Camera 2	/cam_2/image_raw	18.9 Hz · 2 frame gaps
Localization	/mtx/odom	99.9 Hz · no gaps

⏮ ▶ ⏭ **17.9 / 72.1 s** 1× ▾ Click an event or drag the playhead



Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

A Comparison card: the operator stopped this run at 64.7 s. With the trigger list, the stop would have come at 17.9 s.

B Timeline lanes are the mode, speed, trigger events and one lane per camera plus localization.

C Clicking an event moves the map, camera frames and log to that moment.

4 · Trigger list

The conditions that stop the vehicle. Thresholds are first proposals and can be tuned with CAVS.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor `autonomy_demo_0` ▾ **REVIEW** NATURE / MTX stack · 3 cameras **MANUAL**

STOP TRIGGER LIST

ID	TRIGGER	SIGNAL WATCHED	THRESHOLD	ACTION	FIRED IN AUTONOMY_DEMO_0
T1	Camera frame gap	`time between frames, any camera`	> 158 ms (3 frames)	STOP	first at 17.9 s
T2	Camera rate drop	`frames per second, 1 s window`	< 15 Hz	STOP	first at 23.6 s
T3	Camera feed lost	`no frame received`	> 0.5 s	STOP	
T4	Localization stale	`time since last /mtx/odom`	> 0.2 s	STOP	
T5	Autonomy command stale	`/mtx/arbiter_status`	autonomy_cmd_fresh = false	STOP	
T6	Path follower aborted	`controller_server log`	patience exceeded	ALERT	first at 15.5 s
T7	Collision monitor zone	`/mtx/nav/collision_monitor_state`	slowdown or stop zone	ALERT	first at 23.2 s

STOP publishes `/mtx/estop = true` once, from a trigger monitor node running beside the vehicle stack, so a stop does not depend on the network link. The vehicle halts and the safety driver takes over. The interface has no Go or resume control.

ALERT flags the event on the dashboard and in the log. Driving continues.

Every trigger is computed from message timing and small status topics. No image or point-cloud payload is needed to evaluate them.

Dashboard | Path Planning | Faults | Cameras | Run Logs | **Triggers**

A T1–T3 cover the three cameras. T4–T5 cover localization and the autonomy command link.

B T6–T7 are alerts only, since the navigation stack already slows or replans for them.

C The last column comes from replaying the provided bag.

5 · Data budget

Why the interface stays light: triggers are evaluated on the vehicle side, and only small summaries reach the browser.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor `autonomy_demo_0` ▾ **REVIEW** NATURE / MTX stack · 3 cameras **MANUAL**

Vehicle
ROS 2 · NATURE / MTX (or bag playback)
+ trigger monitor node

→ **rosbridge** filtered, throttled topics

CAVS server
hosts the interface
keeps run history

→ **websocket** ≈ 50 kB/s

Browser
any desktop
no install

STAYS ON THE VEHICLE SIDE

3 cameras, raw	`2048 × 1536 mono8 · 19 Hz each`	**≈ 179 MB/s**
Ouster LiDAR point cloud	`10 Hz`	**≈ 63 MB/s**
3D map / point cloud view		**not sent**

SENT TO THE BROWSER

Trigger states	`from the trigger monitor node`	**< 1 kB/s**
Vehicle pose for 2D map	`/mtx/odom throttled to 5 Hz`	**≈ 4 kB/s**
Status + log lines	`/mtx/arbiter_status, /rosout`	**≈ 2 kB/s**
Camera previews	`3 × 320 × 240 JPEG · 1 frame/s`	**≈ 45 kB/s**

≈ 50 kB/s to the browser against **≈ 240 MB/s** of raw camera and LiDAR data on the vehicle.

Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

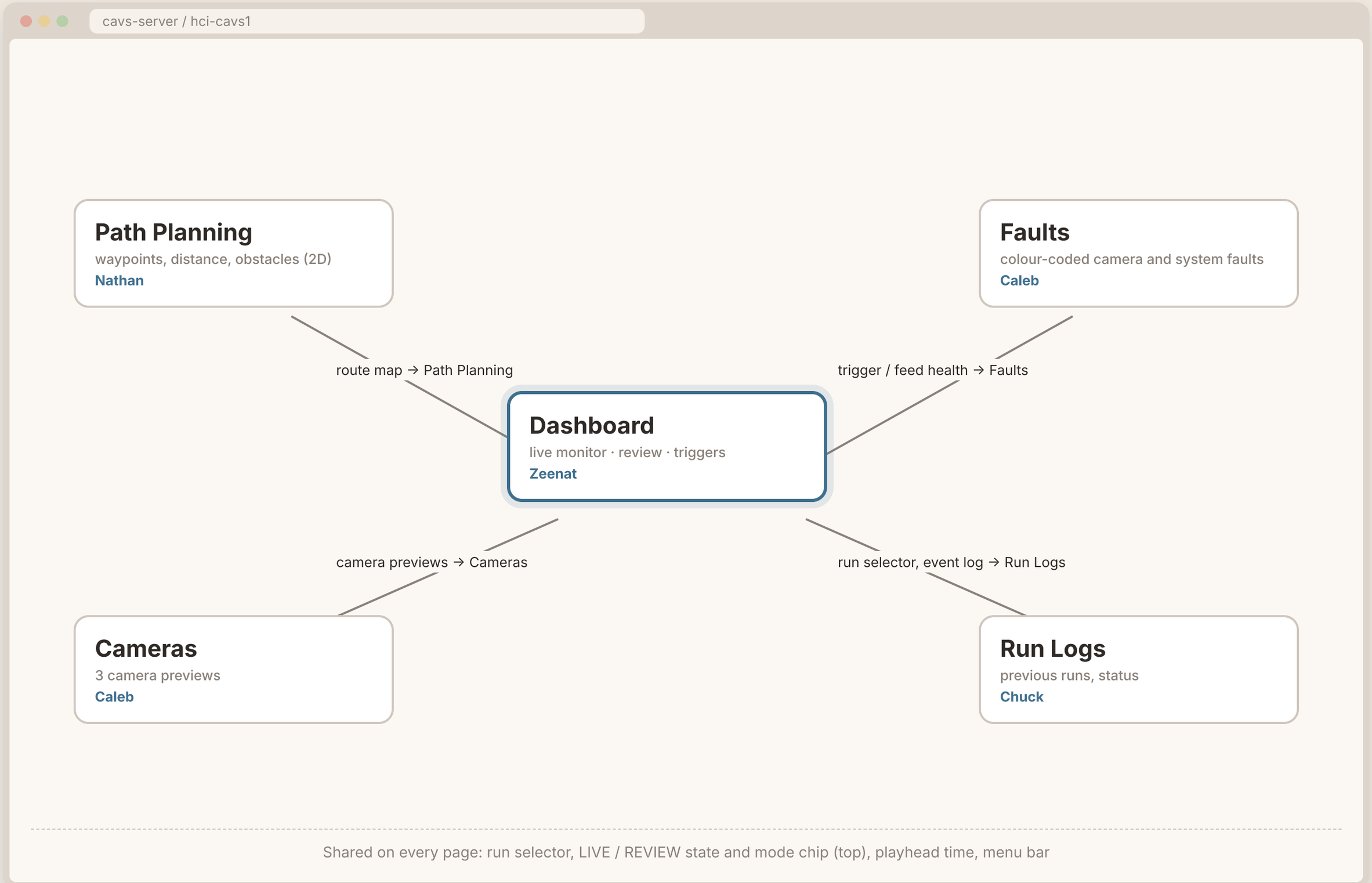
A Raw sizes are measured from the bag: each camera frame is 3.1 MB and each point cloud 6.3 MB.

B Previews and pose are throttled before they leave the server, so several people can watch at once.

C The same layout works with the recorded bag now and with the live vehicle later.

6 · Navigation

Where the dashboard hands off to the other pages in the team's design.



A The dashboard is the hub. Each panel links to the page that owns its detail.

B The selected run and playhead time carry over between pages.